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Inventor(s)	Cross; David I. et al.

Testing a signal path

Abstract

Embodiments of the present disclosure relate generally to testing one or more signal paths. For example, a signal path may include a phase shifter that may impart a phase shift to signals passing through the signal path. Some embodiments may test a phase shift imparted to a signal by the signal path, including the phase shifter. Some embodiments may test the phase shift by comparing the phase of a signal at an input of the signal path with the phase of a signal at the output of the signal path. Some embodiments may test the phase shift by providing a signal at inputs of two phase paths and comparing the phases of signals at the outputs of the signal paths. Some embodiments may further adjust a phase shifter responsive to the test. Related devices, systems and methods are also disclosed.

Inventors: **Cross; David I. (Jessup, MD), Egan; Jonathan D. (Hanover, MD), March; Robert J. (Bel Air, MD), Barkdoll; Ty L. (Columbia, MD)**

Applicant: **Northrop Grumman Systems Corporation** (Falls Church, VA)

Family ID: **89845648**

Assignee: **Northrop Grumman Systems Corporation (Falls Church, VA)**

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Primary Examiner: Tran; Tuan A

Attorney, Agent or Firm: TraskBritt

Background/Summary

FIELD

(1) This description relates, generally, to testing a signal path. More specifically, some examples relate to methods of testing signal paths, devices including one or more signal paths, the devices being capable of testing the one or more signal paths, and/or systems including one or more signal paths, the systems capable of testing the one or more signal paths.

BACKGROUND

(2) A phased-array radar may include an exciter that may generate an oscillating signal e.g., a signal oscillating at radio frequency e.g., between extremely low frequencies (e.g., 3 hertz (Hz)) and tremendously high frequencies (e.g., frequencies in the terahertz (THz) range). The phased-array radar may include a number of antenna elements that may be excited by the oscillating signal to transmit a signal. The phased-array radar may include a respective signal path between each of the number of antenna elements and the exciter. Each of the signal paths may provide the oscillating signal to an antenna element to which the signal path is coupled. Each of the respective signal paths may be capable of independently shifting the phase of the oscillating signal between the exciter and the respective antenna element to steer a beam of the transmitted signal.

(3) The phased-array radar may also receive a signal at antenna elements of the phased-array radar (e.g., a reflection of the transmitted signal). Each of the signal paths may provide the signal, as received at the antenna element to which the signal path is coupled, to an array receiver. Each of the respective signal paths may be capable of independently shifting a phase of the received signal to steer a receiving beam.

SUMMARY

(4) Various embodiments may include a device including signal paths comprising respective front ends for coupling to respective antenna elements and respective back ends for coupling to a manifold. The device may also include a comparator. The comparator may be coupled to a respective front end of a first signal path of the signal paths and coupled to a respective back end of the first signal path, to compare a phase of a first signal from the respective front end of the first signal path with a phase of a second signal from the respective back end of the first signal path. Additionally or alternatively, the comparator may be coupled to the respective front end of the first signal path and coupled to a respective front end of a second signal path of the signal paths, to compare the phase of the first signal from the respective front end of the first signal path with a phase of a third signal from the respective front end of the second signal path. Additionally or alternatively, the comparator may be coupled to the respective back end of the first signal path and coupled to a respective back end of the second signal path, to compare the phase of the second signal from the respective back end of the first signal path with a phase of a fourth signal from the respective back end of the second signal path.

(5) Various embodiments may include a device including a signal path comprising a front end for coupling to an antenna and a back end for coupling to a manifold. The device may also include a comparator coupled to the front end and coupled to the back end. The comparator may compare a phase of a first signal from the front end with a phase of a second signal from the back end.

(6) Various embodiments may include a device including a first signal path comprising a first front

end for coupling to a first antenna and a first back end for coupling to a manifold. The device may also include a second signal path comprising a second front end for coupling to a second antenna and a second back end for coupling to the manifold. The device may also include a comparator coupled to the first front end and coupled to the second front end. The comparator may compare a first signal from the first front end with a second signal from the second front end.
(7) Various embodiments may include a first signal path comprising a first front end for coupling to a first antenna and a first back end for coupling to a manifold. The device may also include a second signal path comprising a second front end for coupling to a second antenna and a second back end for coupling to the manifold. The device may also include a comparator coupled to the back front end and coupled to the second back end. The comparator may compare a first signal from the first back end with a second signal from the second back end.
(8) Various embodiments may include a method. The method may include providing a test signal to an input of a signal path. The method may also include modulating a phase shifter of the signal path. The method may also include comparing a phase of a first signal with a phase of a second signal. The method may also include adjusting the phase shifter responsive to the comparison.
(9) Various embodiments may include a method. The method may include providing a test signal to an input of a signal path. The method may also include modulating a phase shifter of the signal path. The method may also include comparing a phase of the test signal to a phase of an output signal at an output of the signal path. The method may also include adjusting the phase shifter responsive to the comparison.
(10) Various embodiments may include a method. The method may include providing a test signal to an input of a first signal path. The method may also include providing the test signal to an input of a second signal path. The method may also include modulating a first phase shifter of the first signal path. The method may also include comparing a phase of a first output signal at an output of the first signal path with a phase of a second output signal at an output of the second signal path. The method may also include adjusting one or both of the first phase shifter and a second phase shifter of the second signal path responsive to the comparison.
(11) Various embodiments may include a method. The method may include providing a test signal at respective inputs of signal paths. The method may also include activating the signal paths. The method may also include obtaining, at respective comparators at the respective signal paths, phase and amplitude measurements at respective outputs of the signal paths. The method may also include adjusting one or more of the signal paths responsive to the phase and amplitude measurements of the respective signal paths.

Description

BRIEF DESCRIPTION THE DRAWINGS
(1) While this disclosure concludes with claims particularly pointing out and distinctly claiming specific examples, various features and advantages of examples within the scope of this disclosure may be more readily ascertained from the following description when read in conjunction with the accompanying drawings, in which:
(2) FIG. **1** is a functional block diagram illustrating an example device including a signal path.
(3) FIG. **2** is a functional block diagram illustrating an example device including a signal path according to one or more embodiments.
(4) FIG. **3** is a functional block diagram illustrating another example device including a signal path and a signal path according to one or more embodiments.
(5) FIG. **4** is a functional block diagram illustrating yet another example device including a signal path and a signal path according to one or more embodiments.
(6) FIG. **5** is a functional block diagram illustrating yet another example device including a signal

path and a signal path according to one or more embodiments.
(7) FIG. **6** is a functional block diagram illustrating an example system including a multiple signal paths according to one or more embodiments.
(8) FIG. **7** is a functional block diagram illustrating an example comparator according to one or more embodiments.
(9) FIG. **8** is a flowchart of an example method according to one or more embodiments.
(10) FIG. **9** is a flowchart of another example method according to one or more embodiments.
(11) FIG. **10** is a flowchart of yet another example method according to one or more embodiments.
(12) FIG. **11** illustrates a functional block diagram of an example device that may be used to implement various functions, operations, acts, processes, or methods, in accordance with one or more examples.

DETAILED DESCRIPTION

(13) Embodiments of the present disclosure relate generally to testing one or more signal paths. For example, a signal path may include a phase shifter. Some embodiments may test a phase shift imparted to a signal by the signal path, including the phase shifter. Some embodiments may further adjust the phase shifter responsive to the test.
(14) As an example, some embodiments may include a comparator coupled to an input of a signal path (e.g., an input terminal of the signal path) and to an output of the signal path (e.g., to an output terminal of the signal path). The comparator may compare a phase of a signal (e.g., a test signal) at the input of the signal path with a phase of the signal at the output of the signal path. A phase shifter of the signal path may be adjusted such that the signal path cause a specific phase shift (e.g., according to a specification) to signals propagating through the signal path.
(15) As another example, some embodiments may include a comparator coupled to an output of a first signal path (e.g., to an output terminal of the first signal path) and an output of a second signal path (e.g., to an output terminal of the second signal path). A test signal may be provided to respective inputs of the first signal path and the second signal path. The comparator may compare a phase of a signal (e.g., a test signal) as received at the output of the first signal path with the signal as received at the output of the second signal path. A phase shifter of the one or both of the signal paths may be adjusted such that the signal paths cause the same phase shift to signals propagating through the respective signal paths.
(16) As another example, some embodiments may include a number of antenna elements, each of the antenna elements coupled to a respective signal path, each of the signal paths coupled to a respective comparator. All of the signal paths may receive a signal, e.g., a test signal. The signal paths may be tested one at a time by activating the signal paths, one at a time, and measuring an amplitude and/or phase of the signal as received at the output of the active signal path. After testing one or more of the signal paths, one or more of the tested signal paths may be adjusted, e.g., to cause the signal paths to impart uniform phase shift and/or uniform attenuation to signals propagating through the signal paths.
(17) In the present disclosure, the term “signal path” may refer to a path for an electrical signal, e.g., an oscillating signal, e.g., oscillating at radio frequency, between one point and another point. A signal path may include any number of elements, e.g., lines, switches, couplers, amplifiers, and phase shifters.
(18) In the present disclosure, the term “phase shift” may refer to a change in phase of an oscillating signal imparted by a signal path or by a phase shifter of a signal path. For example, a signal path may inherently cause a phase shift to a signal propagating through the signal path, e.g., based on a path length of the signal path. A phase shifter of the signal path may further shift the phase of the signal.
(19) In the present disclosure, the term “attenuation,” and like terms, may refer to a change (e.g., causing a decrease) in an amplitude of a signal. In the present disclosure, the term “gain,” and like terms, may refer to a change (e.g., causing an increase) in an amplitude of a signal. References to

gain and attenuation may be interchangeable.
(20) In the present disclosure, examples are given including switches; the switches may be capable of coupling one line (e.g., an input) to one of two or more other lines (e.g., outputs). Alternative examples may include couplers in place of the switches; the couplers may be capable of redirecting a fraction of energy from one line to one or more other lines.
(21) In the present disclosure, the term “coupled” and derivatives thereof may be used to indicate that two elements co-operate or interact with each other. When an element is described as being “coupled” to or with another element, then the elements may be in direct physical or electrical contact or there may be one or more intervening elements or layers present. In contrast, when an element is described as being “directly coupled” to or with another element, then there are no intervening elements or layers present. It will be understood that when an element is referred to as “coupling” a first element and a second element then it is coupled to the first element and it is coupled to the second element.
(22) In the present disclosure, phased-array radars are given as examples of systems including signal paths. This disclosure is not limited to phased-array radars. Other examples systems that nay include systems for any phased array antenna, including, e.g., communications antennas (e.g., cellular base stations, satellite communications), radio astronomy, radio-frequency altimeters, signals intelligence collection antennas, electronic warfare antennas, radio frequency identification systems, and medical radio frequency ablation antennas.
(23) In the following detailed description, reference is made to the accompanying drawings, which form a part hereof, and in which are shown, by way of illustration, specific examples in which the present disclosure may be practiced. These examples are described in sufficient detail to enable a person of ordinary skill in the art to practice the present disclosure. However, other examples may be utilized, and structural, material, and process changes may be made without departing from the scope of the disclosure.
(24) The illustrations presented herein are not meant to be actual views of any particular method, system, device, or structure, but are merely idealized representations that are employed to describe the examples of the present disclosure. The drawings presented herein are not necessarily drawn to scale. Similar structures or components in the various drawings may retain the same or similar numbering for the convenience of the reader; however, the similarity in numbering does not mean that the structures or components are necessarily identical in size, composition, configuration, or any other property.
(25) The following description may include examples to help enable one of ordinary skill in the art to practice the disclosed examples. The use of the terms “exemplary,” “by example,” and “for example,” means that the related description is explanatory, and though the scope of the disclosure is intended to encompass the examples and legal equivalents, the use of such terms is not intended to limit the scope of an example of this disclosure to the specified components, steps, features, functions, or the like.
(26) It will be readily understood that the components of the examples as generally described herein and illustrated in the drawing could be arranged and designed in a wide variety of different configurations. Thus, the following description of various examples is not intended to limit the scope of the present disclosure, but is merely representative of various examples. While the various aspects of the examples may be presented in drawings, the drawings are not necessarily drawn to scale unless specifically indicated.
(27) Furthermore, specific implementations shown and described are only examples and should not be construed as the only way to implement the present disclosure unless specified otherwise herein. Elements, circuits, and functions may be depicted by block diagram form in order not to obscure the present disclosure in unnecessary detail. Conversely, specific implementations shown and described are only examples and should not be construed as the only way to implement the present disclosure unless specified otherwise herein. Additionally, block definitions and partitioning of

logic between various blocks is an example of a specific implementation. It will be readily apparent to one of ordinary skill in the art that the present disclosure may be practiced by numerous other partitioning solutions. For the most part, details concerning timing considerations and the like have been omitted where such details are not necessary to obtain a complete understanding of the present disclosure and are within the abilities of persons of ordinary skill in the relevant art.

(28) Those of ordinary skill in the art would understand that information and signals may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, and symbols that may be referenced throughout this description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof. Some drawings may illustrate signals as a single signal for clarity of presentation and description. It will be understood by a person of ordinary skill in the art that the signal may represent a bus of signals, wherein the bus may have a variety of bit widths and the present disclosure may be implemented on any number of data signals including a single data signal. A person having ordinary skill in the art would appreciate that this disclosure encompasses communication of quantum information and qubits used to represent quantum information.

(29) The various illustrative logical blocks, modules, and circuits described in connection with the examples disclosed herein may be implemented or performed with a general purpose processor, a special purpose processor, a Digital Signal Processor (DSP), an Integrated Circuit (IC), an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor (may also be referred to herein as a host processor or simply a host) may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices, such as a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. A general-purpose computer including a processor is considered a special-purpose computer while the general-purpose computer executes computing instructions (e.g., software code) related to examples of the present disclosure.

(30) The examples may be described in terms of a process that is depicted as a flowchart, a flow diagram, a structure diagram, or a block diagram. Although a flowchart may describe operational acts as a sequential process, many of these acts can be performed in another sequence, in parallel, or substantially concurrently. In addition, the order of the acts may be re-arranged. A process may correspond to a method, a thread, a function, a procedure, a subroutine, or a subprogram. Furthermore, the methods disclosed herein may be implemented in hardware, software, or both. If implemented in software, the functions may be stored or transmitted as one or more instructions or code on computer-readable media. Computer-readable media includes both computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another.

(31) FIG. **1** is a functional block diagram illustrating an example device **100** including a signal path **102**. Signal path **102** may be an example of a signal path that may be used in one or more embodiments of the present disclosure.

(32) Signal path **102** includes a front end **104** coupled to an antenna element **124** and a back end **106** coupled to a manifold **110**. Signal path **102** may alternately provide a signal to be transmitted from manifold **110** to antenna element **124** and a signal received at antenna element **124** to manifold **110**.

(33) Front end **104** may be, or may include, any point of signal path **102** to the left (from the perspective of FIG. **1**) of phase shifter **114**. Back end **106** may be, or may include, any point of signal path **102** to the right (from the perspective of FIG. **1**) of phase shifter **114**.

(34) Manifold **110** may allow one or more signal paths (e.g., one or more instances of the signal path **102**) to be coupled thereto. For example, multiple signal paths (e.g., including signal path **102**) may be arranged adjacent to one another. Manifold **110** may be coupled to the multiple signal paths. Further, manifold **110** may allow for coupling signal path **102** (and/or other signal paths) to a larger system including e.g., an exciter and/or array receiver.
(35) Signal path **102** includes a variable attenuator **112**. Variable attenuator **112** may adjust an amplitude of a signal propagating through signal path **102**. Variable attenuator **112** may be controlled by a controller (not illustrated in FIG. **1**) to variably adjust the amplitude of the signal propagating through signal path **102**. Variable attenuator **112** may be, or may include, a variable resistor.
(36) Signal path **102** includes a phase shifter **114**. Phase shifter **114** may adjust the phase of a signal propagating through signal path **102**. Phase shifter **114** may be, or may include, one or more selectable delay elements or delay paths. Phase shifter **114** may be controlled by a controller (not illustrated in FIG. **1**) to variably adjust the phase of the signal propagating through signal path **102**.
(37) Signal path **102** includes switch **116** and switch **122**. Switch **116** and switch **122** may be controlled to operate together to cause a signal from manifold **110** to propagate through amplifier **118** and a signal from antenna element **124** to propagate through amplifier **120**. Amplifier **118** may be a high-power amplifier to amplify a signal from manifold **110** to be transmitted at antenna element **124**. Amplifier **120** may be a low-noise amplifier to amplify a signal received at antenna element **124**.
(38) Antenna element **124** may be an antenna or a radiator to be excited by an oscillating signal (to transmit) or by an electromagnetic field (to receive). Exciting antenna element **124** with an oscillating signal (e.g., provided by signal path **102**) may generate a transmitted signal transmitted at antenna element **124** as an electromagnetic field. Also, an electromagnetic field impinging on antenna element **124** may excite antenna element **124**. Antenna element **124**, so excited, may generate a received signal that may propagate through signal path **102**.
(39) In some cases, signal path **102** (including variable attenuator **112**, phase shifter **114**, switch **116**, amplifier **118**, amplifier **120**, and switch **122**), may be included on a radio-frequency integrated circuit **126** (RFIC **126**). In some embodiments, manifold **110** may, or may not, be considered part of signal path **102** (though in FIG. **1**, manifold **110** is illustrated as included in signal path **102**). In some embodiments, manifold **110** may or may not be included on RFIC **126** (though in FIG. **1**, manifold **110** is illustrated as included in RFIC **126**).
(40) FIG. **2** is a functional block diagram illustrating an example device **200** including a signal path **202** according to one or more embodiments. Signal path **102** of FIG. **1** may be an example of signal path **202** of FIG. **2**. Device **200** may compare a signal from a front end **204** of signal path **202** (e.g., illustrated by the dashed line between switch **222** and comparator **208**) with a signal from a back end **206** of signal path **202** (e.g., illustrated by the dashed line between switch **228** and comparator **208**). Device **200** may, responsive to the comparison, determine a status of signal path **202** and/or adjust signal path **202**.
(41) In the present disclosure, elements of some drawings (e.g., FIG. **1** through FIG. **6**) may be the same as, be substantially similar to, and/or function substantially the same as, elements of other drawings. Further, a reference number having the same last two digits as a corresponding reference number in another drawing, may indicate that elements referenced by the respective reference numbers are substantially the same, absent explicit description to the contrary. For example, manifold **210** of FIG. **2** may be the same as, or substantially similar to manifold **110** of FIG. **1**.
(42) In the present disclosure, dashed lines may be used to illustrate signals, e.g., test signals and/or signals to be compared e.g., at a comparator. Such signals may traverse signal paths e.g., signaling lines. Thus, the dashed lines may also represent signal paths. Additionally, solid lines between elements may also illustrate signals and/or signal paths. Some of the solid lines between elements may illustrate signal paths that may be used during operation of the various devices and systems

described herein. For example, the solid line between antenna element **224** and switch **222** may illustrate a signal path used during operation of device **200**. In contrast, some of the dashed lines may illustrate signal paths that may be used during testing of the various devices and systems described herein. For example, the dashed line between switch **222** and comparator **208** and the dashed line between switch **228** and comparator **208** may be used during testing. However, some of the solid lines and some of the dashed lines may represent signal paths used during both operation and testing. For example, test signal **232** may traverse the solid line between antenna element **224** and switch **222**.

(43) Device **200** includes signal path **202**. Signal path **202** includes front end **204** for coupling to antenna element **224** and back end **206** for coupling to manifold **210**. Device **200** includes a comparator **208** coupled to front end **204** and coupled to back end **206**. Comparator **208** may compare a phase (and/or amplitude) of a first signal from front end **204** with a phase (and/or amplitude) of a second signal from back end **206**.

(44) Comparator **208** may be coupled at any point of front end **204** of signal path **202**. For example, comparator **208** may be coupled anywhere to the left (from the perspective of FIG. **2**) of a phase shifter of signal path **202** (e.g., phase shifter **114** of FIG. **1**, phase shifter not shown in FIG. **2**). Comparator **208** may be coupled at any point of back end **206** of signal path **202** and/or to switch **228**. For example, comparator **208** may be coupled anywhere to the right (from the perspective of FIG. **2**) of a phase shifter of signal path **202**, (e.g., phase shifter **114** of FIG. **1**, phase shifter not shown in FIG. **2**). Additionally or alternatively, comparator **208** may be coupled to switch **228** or to another manifold or switch to the right (from the perspective of FIG. **2**) of signal path **202**.

(45) Device **200** may also include a switch **228**. Switch **228** may variously couple manifold **210** to other elements e.g., to other manifolds and/or to elements of a larger system e.g., an array receiver and/or array exciter.

(46) Device **200** may also include a controller **230**. Controller **230** may control operation of a phase shifter (e.g., phase shifter **114** of FIG. **1**, not illustrated in FIG. **2**) of signal path **202**. Controller **230** may also adjust the phase shifter. Controller **230** may also be capable of adjusting a variable attenuator (e.g., variable attenuator **112** of FIG. **1**) of signal path **202**. Further, controller **230** may also activate or deactivate signal path **202**.

(47) Device **200** may also include a test-signal provider **240**. Test-signal provider **240** may provide one or more test signals (e.g., test signal **234** and/or test signal **236**) to one or more locations of device **200**.

(48) As an example of contemplated operations of device **200**, antenna element **224** may receive a test signal **231**. Test signal **231** may be a signal propagating as an electromagnetic field. Antenna element **224** may be excited responsive to the test signal **231**. Test signal **231** may have been transmitted by another antenna, e.g., in an antenna range to test a system, e.g., a system including signal path **202**.

(49) Continuing the example of contemplated operations, antenna element **224** may provide test signal **232**, which test signal **232** may be test signal **231** as received at antenna element **224**, to signal path **202** at front end **204**, e.g., at switch **222**. Test signal **232** may propagate through signal path **202** from front end **204** to back end **206**, e.g., to manifold **210**. Manifold **210** may provide a signal, e.g., test signal **232** as test signal **232** was received at manifold **210**, to switch **228**. Switch **228** may provide a signal, e.g., test signal **232** as test signal **232** was received at switch **228** to comparator **208**. Alternatively, although not shown in FIG. **2**, comparator **208** may be coupled to manifold **210** directly (e.g., rather than through the switch **228** as illustrated in FIG. **2**) and may receive a signal, e.g., test signal **232** as test signal **232** was received at manifold **210**.

(50) As another example of contemplated operations of device **200**, test-signal provider **240** may provide a test signal **234** to signal path **202**, e.g., at switch **222**. Test signal **234** may replicate a signal that would be received at antenna element **224**. Test signal **234** may propagate through signal

path **202** from front end **204** to back end **206**, e.g., to manifold **210**. Manifold **210** may provide a signal, e.g., test signal **234**, as test signal **234** was received at manifold **210**, to switch **228**. Switch **228** may provide a signal, e.g., test signal **234**, as test signal **234** was received at switch **228** to comparator **208**. Alternatively, comparator **208** may be coupled to manifold **210** directly (e.g., rather than through the switch **228** as illustrated in FIG. **2**) and may receive a signal, e.g., test signal **234** as test signal **234** was received at manifold **210**.

(51) As another example of contemplated operations of device **200**, test-signal provider **240** may provide a test signal **236** to signal path **202** e.g., at switch **228** (or manifold **210**). Test signal **236** may be a signal suitable for transmission at antenna element **224**. Test signal **236** may propagate through signal path **202** from back end **206** to front end **204**, e.g., to switch **222**. Switch **222** may provide a signal, e.g., test signal **236**, as test signal **236** was received at switch **222**, to comparator **208**.

(52) As another example of contemplated operations of device **200**, switch **228** (or manifold **210**) may receive test signal **238**. Test signal **238** may be a signal suitable for transmission at antenna element **224**. Test signal **238** may be provided to switch **228** from another manifold and/or another element of an antenna system. For example, test signal **238** may be generated by an array exciter. Test signal **238** may propagate through signal path **202** from back end **206** to front end **204**, e.g., to switch **222**. Switch **222** may provide a signal, e.g., test signal **236**, as test signal **236** was received at switch **222**, to comparator **208**.

(53) According to any of the contemplated examples of operations above, or another example, comparator **208** may be coupled to front end **204** and to back end **206**. Comparator **208** may compare the phase and/or amplitude of a signal from front end **204** with the phase and/or amplitude of a signal from back end **206**. Comparator **208** and/or controller **230** may have information regarding a nominal phase shift and/or nominal attenuation of signal path **202**, e.g., according to a specification of signal path **202**. Additionally or alternatively, comparator **208** and/or controller **230** may have information regarding a threshold for phase shift and/or a threshold for attenuation of signal path **202**.

(54) In some embodiments, responsive to the comparison of the signal at front end **204** to the signal at back end **206**, comparator **208** and/or controller **230** may determine a status (e.g., an operability status) of signal path **202**. For example, based on a determination that a phase shift imparted to the test signal by signal path **202** and/or an attenuation imparted to the test signal by signal path **202** are not within respective thresholds (e.g., thresholds of the specification of signal path **202**), comparator **208** and/or controller **230** may determine that signal path **202** is inoperative. In some embodiments, responsive to a determination that signal path **202** is inoperative, device **200** may determine not to use signal path **202**, additionally or alternatively, controller **230** may disable signal path **202**.

(55) Additionally or alternatively, comparator **208** and/or controller **230** may adjust a phase shift and/or an attenuation imparted to signals by signal path **202** to cause the phase shift and/or attenuation to be within the threshold and/or to be closer to the nominal phase shift and or the nominal attenuation. For example, controller **230** may control a phase shifter (e.g., phase shifter **114** of FIG. **1**, phase shifter not shown in FIG. **2**) to adjust a phase shift of signal path **202** to cause the phase shift of signal path **202** to be substantially similar to the nominal phase shift of signal path **202**. Additionally or alternatively, controller **230** may control a variable attenuator (e.g., variable attenuator **112** of FIG. **1**) to adjust an attenuation of signal path **202** to cause the attenuation of signal path **202** to be substantially similar to the nominal attenuation of signal path **202**.

(56) As an example of comparing a test signal from front end **204** to a test signal from back end **206**, the test signal from front end **204** and the test signal from back end **206** may be combined e.g., at coupler or summer. A degree to which the test signal from front end **204** and the test signal from back end **206** constructively or destructively interfere may be measured. For example, the

combined signals may be input in a rectifier and an amplitude of the rectified signal may be indicative of the degree of constructive or destructive interference of the test signal from front end **204** with the test signal from back end **206**.
(57) According to example contemplated operations, controller **230** may modulate the phase shifter of signal path **202** e.g., controller **230** may cause the phase shifter to cycle through phases of a phase range. When the test signal from front end **204** is 180° out of phase with the test signal from back end **206** the test signal from front end **204** and the test signal from back end **206** will exhibit a greatest degree of destructive interference one with another. Thus, a phase of the phase range that causes a greatest degree of destructive interference between the test signal from front end **204** and the test signal from back end **206** may correspond to a 180° phase shift of the phase shifter.
(58) Comparator **208** and/or controller **230** may determine a phase of phase shifter at which the test signal from front end **204** and the test signal from back end **206** exhibit a greatest degree of destructive interference and controller **230** may set the phases of the phase shifter such that the determined phase corresponds to a 180° phase shift. Alternatively, comparator **208** and/or controller **230** may determine a phase of phase shifter at which the test signal from front end **204** and the test signal from back end **206** exhibit a greatest degree of constructive interference and controller **230** may set the phases of the phase shifter such that the determined phase corresponds to a 0° phase shift.
(59) Additionally or alternatively, comparator **208** and/or controller **230** may determine an attenuation of signal path **202** and may adjust the attenuation of signal path **202**, e.g., such that the attenuation is within a threshold or close to a nominal attenuation. Comparator **208** and/or controller **230** may determine the attenuation by observing the degrees of constructive and destructive interference.
(60) In some embodiments, signal path **202** may be arranged on RFIC **226**. Additionally or alternatively, comparator **208**, switch **228**, controller **230**, and/or test-signal provider **240** may also be arranged on RFIC **226**.
(61) FIG. **3** is a functional block diagram illustrating another example device **300** including a signal path **302***a* and a signal path **302***b* according to one or more embodiments. Signal path **102** of FIG. **1** may be an example of signal path **302***a* and of signal path **302***b*. Device **300** may compare a signal from a front end **304***a* of signal path **302***a* with a signal from front end **304***b* of signal path **302***b*. Device **300** may, responsive to the comparison, determine a status of signal path **302***a* and/or signal path **302***b* and/or adjust signal path **302***a* and/or signal path **302***b*.
(62) Device **300** includes signal path **302***a*, which signal path **302***a* includes front end **304***a* for coupling to an antenna element **324***a* and a back end **306***a* for coupling to manifold **310**. Device **300** includes signal path **302***b*, which signal path **302***b* includes front end **304***b* for coupling to an antenna element **324***b* and a back end **306***b* for coupling to manifold **310**. Device **300** includes a comparator **308** coupled to front end **304***a* and coupled to front end **304***b*. Comparator **308** may compare a first signal from front end **304***a* with a second signal from front end **304***b*.
(63) Comparator **308** may be the same as, be substantially similar to, and/or function substantially the same as, comparator **208** of FIG. **2**. However, unlike comparator **208** of FIG. **2**, comparator **308** may be coupled to front end **304***a* and coupled to front end **304***b*. Similar to comparator **208** of FIG. **2**, comparator **308** may be coupled at any point of front end **304***a* of signal path **302***a* and at any point of front end **304***b* of signal path **302***b*.
(64) Controller **330** may be the same as, be substantially similar to, and/or function substantially the same as, controller **230** of FIG. **2**. However, controller **330** may control phase shifters and/or variable attenuators of signal path **302***a* and signal path **302***b*.
(65) As an example of contemplated operations of device **300**, test-signal provider **340** may provide a test signal **336** to signal path **302***a* and signal path **302***b* e.g., at switch **328** (or manifold **310**). Test signal **336** may be a signal suitable for transmission at antenna element **324***a* and antenna element **324***b*. Test signal **336** may propagate through signal path **302***a* from back end **306***a* to front

end **304***a*, e.g., to switch **322***a* and through signal path **302***b* from back end **306***b* to front end **304***b*, e.g., to switch **322***b*. Switch **322***a* and switch **322***b* may provide a signal, e.g., test signal **336** as test signal **336** was received at switch **322***a* and switch **322***b* respectively, to comparator **308**.
(66) As another example of contemplated operations of device **300**, switch **328** (or manifold **310**) may receive test signal **338**. Test signal **338** may be a signal suitable for transmission at antenna element **324***a* and antenna element **324***b*. Test signal **338** may be provided to switch **328** from another manifold and/or another element of an antenna system. For example, test signal **338** may be generated by an array exciter. Test signal **338** may propagate through signal path **302***a* from back end **306***a* to front end **304***a*, e.g., to switch **322***a* and through signal path **302***b* from back end **306***b* to front end **304***b*, e.g., to switch **322***b*. Switch **322***a* and switch **322***b* may provide a signal, e.g., test signal **338** as test signal **338** was received at switch **322***a* and switch **322***b* respectively, to comparator **308**.
(67) According to either of the contemplated examples of operations above, or another example, comparator **308** may be coupled to front end **304***a* and to front end **304***b*. Comparator **308** may compare the phase and/or amplitude of a signal from front end **304***a* with the phase and/or amplitude of a signal from front end **304***b*.
(68) In some embodiments, responsive to the comparison of the signal at front end **304***a* to the signal at front end **304***b*, comparator **308** and/or controller **330** may determine a status (e.g., an operability status) of one or both of signal path **302***a* and signal path **302***b*. For example, based on a determination that a phase shift imparted to the test signal by signal path **302***b* and/or an attenuation imparted to the test signal by signal path **302***b* are not within respective thresholds (e.g., thresholds of the specification of signal path **302***b*) from the test signal at front end **304***a*, comparator **308** and/or controller **330** may determine that signal path **302***b* is inoperative. In some embodiments, responsive to a determination that signal path **302***b* is inoperative, device **300** may determine not to use signal path **302***b*, additionally or alternatively, controller **330** may disable signal path **302***b*.
(69) Additionally or alternatively, comparator **308** and/or controller **330** may adjust a phase shift and/or an attenuation of signal path **302***b* to cause the signal at front end **304***b* to be within the threshold range of the signal at front end **304***a*. For example, controller **330** may control a phase shifter (e.g., phase shifter **114** of FIG. **1**, phase shifter not shown in FIG. **3**) to adjust a phase shift imparted by signal path **302***b* to be substantially the same as the phase shift imparted by signal path **302***a*. Additionally or alternatively, controller **330** may control a variable attenuator (e.g., variable attenuator **112** of FIG. **1**) to adjust an attenuation imparted by signal path **302***b* to cause the attenuation of signal path **302***b* to be substantially the same as the attenuation imparted by signal path **302***a*.
(70) As an example of comparing a test signal from front end **304***a* to a test signal from front end **304***b*, the test signal from front end **304***a* and the test signal from front end **304***b* may be combined e.g., at coupler or summer. A degree to which the test signal from front end **304***a* and the test signal from front end **304***b* constructively or destructively interfere may be measured. For example, the combined signals may be input in a rectifier and an amplitude of the rectified signal may be indicative of the degree of constructive or destructive interference of the test signal from front end **304***a* and the test signal from front end **304***b*.
(71) According to example contemplated operations, controller **330** may modulate the phase shifter of signal path **302***b* e.g., controller **330** may cause the phase shifter to cycle through phases of a phase range. When the test signal from front end **304***b* is 180° out of phase with the test signal from front end **304***a* the test signal from front end **304***a* and the test signal from front end **304***b* will exhibit a greatest degree of destructive interference one with another. Thus, a phase of the phase range that causes a greatest degree of destructive interference between the test signal from front end **304***a* and the test signal from front end **304***b* may correspond to a 180° difference between phases of the phase shifter of signal path **302***a* and the phase shifter of signal path **302***b*.

(72) Comparator **308** and/or controller **330** may determine a phase of the phase shifter of signal path **302**_b_ relative to a phase of the phase shifter of signal path **302**_a_ at which the test signal from front end **304**_a_ and the test signal from front end **304**_b_ exhibit a greatest degree of destructive interference. Controller **330** may set the phases of the phase shifter of signal path **302**_b_ such that the determined phase corresponds to a 180° phase difference between the phase shifter of signal path **302**_b_ and the phase shifter of signal path **302**_a_. Alternatively, comparator **308** and/or controller **330** may determine a phase of phase shifter at which the test signal from front end **304**_a_ and the test signal from front end **304**_b_ exhibit a greatest degree of constructive interference and controller **330** may set the phases of the phase shifter of signal path **302**_b_ such that the determined phase corresponds to a 0° phase difference between the between the phase shifter of signal path **302**_b_ and the phase shifter of signal path **302**_a_.
(73) Additionally or alternatively, comparator **308** and/or controller **330** may determine a difference between attenuation of signal path **302**_a_ attenuation of signal path **302**_b_. Comparator **308** and/or controller **330** may adjust the attenuation of signal path **302**_a_ and/or signal path **302**_b_ e.g., such that the attenuation of signal path **302**_b_ is within a threshold range from the attenuation of signal path **302**_a_. Comparator **308** and/or controller **330** may determine the attenuation by observing the degrees of constructive and destructive interference between the signal at front end **304**_a_ and the signal at front end **304**_b_.
(74) In some embodiments, signal path **302**_a_ and signal path **302**_b_ may be arranged on RFIC **326**. In other embodiments, signal path **302**_a_ and signal path **302**_b_ may be arranged on separate RFICs (though not illustrated as such in FIG. **3**). In such embodiments, signal path **302**_a_ and signal path **302**_b_ may nevertheless be coupled at a manifold **310**.
(75) FIG. **4** is a functional block diagram illustrating yet another example device **400** including a signal path **402**_a_ and a signal path **402**_b_ according to one or more embodiments. Signal path **102** of FIG. **1** may be an example of signal path **402**_a_ and of signal path **402**_b_. Device **400** may compare a signal from a back end **406**_a_ of signal path **402**_a_ with a signal from back end **406**_b_ of signal path **402**_b_. Device **400** may, responsive to the comparison, determine a status of signal path **402**_a_ and/or signal path **402**_b_ and/or adjust signal path **402**_a_ and/or signal path **402**_b_.
(76) Device **400** includes signal path **402**_a_, which signal path **402**_a_ includes front end **404**_a_ for coupling to an antenna element **424**_a_ and a back end **406**_a_ for coupling to manifold **410**. Device **400** includes signal path **402**_b_, which signal path **402**_b_ includes front end **404**_b_ for coupling to an antenna element **424**_b_ and a back end **406**_b_ for coupling to manifold **410**. Device **400** includes a comparator **408** coupled to back end **406**_a_ and coupled to back end **406**_b_. Comparator **408** may compare a first signal from back end **406**_a_ with a second signal from back end **406**_b_.
(77) Comparator **408** may be the same as, be substantially similar to, and/or function substantially the same as, comparator **308** of FIG. **3** and/or comparator **208** of FIG. **2**. However, unlike comparator **308** and comparator **208**, comparator **408** may be coupled to back end **406**_a_ and coupled to back end **406**_b_. Similar to comparator **208**, comparator **408** may be coupled at any point of back end **406**_a_ of signal path **402**_a_ and at any point of back end **406**_b_ of signal path **402**_b_.
(78) As an example of contemplated operations of device **400**, antenna element **424**_a_ may receive a test signal **431**_a_ and antenna element **424**_b_ may receive a test signal **431**_b_. Test signal **431**_a_ and test signal **431**_b_ may be a signal propagating as an electromagnetic field, e.g., as received at antenna element **424**_a_ and antenna element **424**_b_ respectively. Antenna element **424**_a_ and antenna element **424**_b_ may be excited by the test signal. Test signal **431**_a_ and test signal **431**_b_ may have been transmitted by another antenna, e.g., in an antenna range to test a system, e.g., a system including signal path **402**_a_ and signal path **402**_b_.
(79) Continuing the example of contemplated operations, antenna element **424**_a_ may provide test signal **432**_a_, which test signal **432**_a_ may be test signal **431**_a_ as test signal **431**_a_ was received at antenna element **424**_a_, to signal path **402**_a_ at front end **404**_a_, e.g., at switch **422**_a_. Additionally, antenna element **424**_b_ may provide test signal **432**_b_, which test signal **432**_a_ may be test signal **431**_b_

as test signal **431***b* was received at antenna element **424***b*, to signal path **402***b* at front end **404***b*, e.g., at switch **422***b*. Test signal **432***a* may propagate through signal path **402***a* from front end **404***a* to back end **406***a*, e.g., to manifold **410**. Test signal **432***b* may propagate through signal path **402***b* from front end **404***b* to back end **406***b*, e.g., to manifold **410**. Manifold **410** may provide a first signal, e.g., test signal **432***a* as test signal **432***a* was received at manifold **410**, and a second signal, e.g., test signal **432***b* as test signal **432***b* was received at manifold **410**, to switch **428**. Switch **428** may provide the first signal and the second signal as they were respectively received at switch **428** (e.g., combined) to comparator **408**. Alternatively, comparator **408** may be coupled to manifold **410** and may receive first and second signals as they were received at manifold **410** (e.g., combined).
(80) As another example of contemplated operations of device **400**, test-signal provider **440** may provide a test signal **434***a* to signal path **402***a*, e.g., at switch **422***a* and a test signal **434***b* to signal path **402***b*, e.g., at switch **422***b*. Test signal **434***a* and test signal **434***b* may replicate a signal that would be received at antenna element **424***a* and antenna element **424***b* respectively. Test signal **434***a* may propagate through signal path **402***a* from front end **404***a* to back end **406***a*, e.g., to manifold **410**. Test signal **434***b* may propagate through signal path **402***b* from front end **404***b* to back end **406***b*, e.g., to manifold **410**. Manifold **410** may provide a signal, e.g., test signal **434***a* and test signal **434***b* as test signal **434***a* and test signal **434***b* were received at manifold **410**, to switch **428**. Switch **428** may provide the signal as the signal was received at switch **428** (e.g., which signal may include a combination of test signal **434***a* and test signal **434***b*) to comparator **408**. Alternatively, comparator **408** may receive be coupled to manifold **410** and may receive the signal e.g., including test signal **434***a* and test signal **434***b* as they were received at manifold **410** (e.g., combined).
(81) According to either of the contemplated examples of operations above, or another example, comparator **408** may be coupled to back end **406***a* and to back end **406***b*. Comparator **408** may compare the phase and/or amplitude of a signal from back end **406***a* with the phase and/or amplitude of a signal from back end **406***b*.
(82) In some embodiments, responsive to the comparison of the signal at **406***a* to the signal at back end **406***b*, comparator **408** and/or controller **430** may determine a status (e.g., an operability status) of one or both of signal path **402***a* and signal path **402***b*. For example, based on a determination that a phase shift imparted to the test signal by signal path **402***b* and/or an attenuation imparted to the test signal by signal path **402***b* are not within respective thresholds (e.g., thresholds of the specification of signal path **402***b*) from the test signal at back end **406***a*, comparator **408** and/or controller **430** may determine that signal path **402***b* is inoperative. In some embodiments, responsive to a determination that signal path **402***b* is inoperative, device **400** may determine not to use signal path **402***b*, additionally or alternatively, controller **430** may disable signal path **402***b*.
(83) Additionally or alternatively, comparator **408** and/or controller **430** may adjust a phase shift and/or an attenuation of signal path **402***b* to cause the signal at back end **406***b* to be within the threshold range of the signal at back end **406***a*. For example, controller **430** may control a phase shifter (e.g., phase shifter **114** of FIG. **1**, phase shifter not shown in FIG. **4**) to adjust a phase shift of signal path **402***b* to cause the phase shift imparted by signal path **402***b* to be substantially similar to the phase shift imparted by signal path **402***a*. Additionally or alternatively, controller **430** may control a variable attenuator (e.g., variable attenuator **112** of FIG. **1**) to adjust an attenuation imparted by signal path **402***b* to cause the attenuation of signal path **402***b* to be substantially similar to the attenuation imparted by signal path **402***a*.
(84) As an example of comparing a test signal from back end **406***a* to a test signal from back end **406***b*, the test signal from back end **406***a* and the test signal from back end **406***b* may be combined e.g., manifold **410**. A degree to which the test signal from back end **406***a* and the test signal from back end **406***b* constructively or destructively interfere may be measured. For example, the combined signals may be input in a rectifier and an amplitude of the rectified signal may be indicative of the degree of constructive or destructive interference of the test signal from back end

406*a* and the test signal from back end **406***b*.
(85) According to example contemplated operations, controller **430** may modulate the phase shifter of signal path **402***b* e.g., controller **430** may cause the phase shifter to cycle through phases of a phase range. When the test signal from back end **406***b* is 180° out of phase with the test signal from back end **406***a* the test signal from back end **406***a* and the test signal from back end **406***b* will exhibit a greatest degree of destructive interference one with another. Thus, a phase of the phase range that causes a greatest degree of destructive interference between the test signal from back end **406***a* and the test signal from back end **406***b* may correspond to a 180° difference between phases of the phase shifter of signal path **402***a* and the phase shifter of signal path **402***b*.
(86) Comparator **408** and controller **430** may determine a phase of the phase shifter of signal path **402***b* relative to a phase of the phase shifter of signal path **402***a* at which the test signal from back end **406***a* and the test signal from back end **406***b* exhibit a greatest degree of destructive interference. Controller **430** may set the phases of the phase shifter of signal path **402***b* such that the determined phase corresponds to a 180° phase difference between the phase shifter of signal path **402***b* and the phase shifter of signal path **402***a*. Alternatively, comparator **408** and controller **430** may determine a phase of phase shifter at which the test signal from back end **406***a* and the test signal from back end **406***b* exhibit a greatest degree of constructive interference and controller **430** may set the phases of the phase shifter of signal path **402***b* such that the determined phase corresponds to a 0° phase difference between the between the phase shifter of signal path **402***b* and the phase shifter of signal path **402***a*.
(87) Additionally or alternatively, comparator **408** and controller **430** may determine a difference between attenuation of signal path **402***a* attenuation of signal path **402***b*. Comparator **408** and controller **430** may adjust the attenuation of signal path **402***a* and/or signal path **402***b* e.g., such that the attenuation of signal path **402***b* is within a threshold range from the attenuation of signal path **402***a*. Comparator **408** and controller **430** may determine the attenuation by observing the degrees of constructive and destructive interference between the signal at back end **406***a* and the signal at back end **406***b*.
(88) FIG. **5** is a functional block diagram illustrating yet another example device **500** including a signal path **502***a* and a signal path **502***b* according to one or more embodiments. Signal path **102** of FIG. **1** may be an example of signal path **502***a* and of signal path **502***b*. Device **500** may include one or more of comparator **508***a*, comparator **508***b*, and comparator **508***c* to compare various signals from various points of signal path **502***a* and signal path **502***b*. Device **500** may, responsive to the comparison, determine a status of signal path **502***a* and/or signal path **502***b* and/or adjust signal path **502***a* and/or signal path **502***b*.
(89) Device **500** may include comparator **508***a*, which comparator **508***a* may be the same as, be substantially similar to, and/or function substantially the same as, comparator **208** of FIG. **2**. For example, comparator **508***a* may compare a signal from front end **504***a* with a signal from back end **506***a*. Controller **530** may adjust a phase shifter of signal path **502***a* responsive to the comparison.
(90) Device **500** may include comparator **508***b*, which comparator **508***b* may be the same as, be substantially similar to, and/or function substantially the same as, comparator **308** of FIG. **3**. For example, comparator **508***b* may compare a signal from front end **504***a* with a signal from front end **504***b*. Controller **530** may adjust a phase shifter of either of signal path **502***a* or signal path **502***b* responsive to the comparison.
(91) Device **500** may include comparator **508***c*, which comparator **508***c* may be same as, be substantially similar to, and/or function substantially the same as, comparator **408** of FIG. **4**. For example, comparator **508***c* may compare a signal from back end **506***a* with a signal from back end **506***b*. Controller **530** may adjust a phase shifter of either of signal path **502***a* or signal path **502***b* responsive to the comparison.
(92) FIG. **6** is a functional block diagram illustrating an example system **600** including a multiple signal paths according to one or more embodiments. Signal path **102** of FIG. **1** may be an example

of signal path **602***a*, of signal path **602***b*, of signal path **602***c*, and of signal path **602***d*. System **600** may include one or more of comparator **608***a*, comparator **608***b*, comparator **608***c*, comparator **608***d*, comparator **608***e*, and comparator **608***f* to compare various signals from various points of signal path **602***a*, signal path **602***b*, signal path **602***c*, and signal path **602***d*. System **600** may, responsive to the comparison, determine a status of signal path **602***a*, signal path **602***b*, signal path **602***c*, and signal path **602***d* and/or adjust signal path **602***a*, signal path **602***b*, signal path **602***c*, and signal path **602***d*.
(93) System **600** includes signal path **602***a* and signal path **602***b* and signal path **602***c* and signal path **602***d* which may same as, be substantially similar to, and/or function substantially the same as, signal path **502***a* of FIG. **5** and signal path **502***b* of FIG. **5** respectively. Signal path **602***a* and signal path **602***b* are coupled to manifold **610***a* and switch **628***a*, which manifold **610***a* and switch **628***a* may be same as, be substantially similar to, and/or function substantially the same as, manifold **510** of FIG. **5** and switch **528** respectively. Signal path **602***c* and signal path **602***d* are similarly coupled to manifold **610***b* and switch **628***b*, which manifold **610***b* and switch **628***b* may be same as, be substantially similar to, and/or function substantially the same as, manifold **510** and switch **528** respectively. System **600** may include controller **630***a* to control a phase shifter and/or variable attenuator of signal path **602***a* and signal path **602***b* respectively. System **600** may include controller **630***b* to control a phase shifter and/or variable attenuator of signal path **602***c* and signal path **602***d* respectively. Controller **630***a* and controller **630***b* may be same as, be substantially similar to, and/or function substantially the same as, controller **530** of FIG. **5**. System **600** includes test-signal provider **640***a*, which test-signal provider **640***a* may provide test signal **634***a* to signal path **602***a*, test signal **634***b* to signal path **602***b*, and test signal **636***a* to switch **628***a*. System **600** includes test-signal provider **640***b*, which test-signal provider **640***b* may provide test signal **634***c* to signal path **602***c*, test signal **634***d* to signal path **602***d*, and test signal **636***b* to switch **628***b*. Test-signal provider **640***a* and test-signal provider **640***b* may be same as, be substantially similar to, and/or function substantially the same as, test-signal provider **540** of FIG. **5**. Signal path **602***a*, signal path **602***b*, manifold **610***a*, switch **628***a*, and test-signal provider **640***a* may be arranged on RFIC **626***a*. Signal path **602***c*, signal path **602***d*, manifold **610***b*, switch **628***b*, and test-signal provider **640***b* may be arranged on RFIC **626***b*. RFIC **626***a* and RFIC **626***b* may be same as, be substantially similar to, and/or function substantially the same as, RFIC **526** of FIG. **5**.
(94) Additionally, system **600** may include oscillator **642***a*, which oscillator **642***a* may generate an oscillating signal and provide the oscillating signal to test-signal provider **640***a*. Test-signal provider **640***a* may provide the oscillating signal generated by oscillator **642***a* to signal path **602***a*, signal path **602***b*, and/or switch **628***a*, as test signal **634***a*, test signal **634***b*, and/or test signal **636***a* respectively. Oscillator **642***a* may be arranged on RFIC **626***a*.
(95) Additionally, system **600** may include oscillator **642***b*, which oscillator **642***b* may generate an oscillating signal and provide the oscillating signal to test-signal provider **640***b*. Test-signal provider **640***b* may provide the oscillating signal generated by oscillator **642***b* to signal path **602***c*, signal path **602***d*, and/or switch **628***b*, as test signal **634***c*, test signal **634***d*, and/or test signal **636***b* respectively. Oscillator **642***b* may be arranged on RFIC **626***b*.
(96) Additionally, system **600** includes array receiver/exciter **648**. Array receiver/exciter **648** may generate an oscillating signal and provide the oscillating signal to test-signal provider **640***a* and test-signal provider **640***b* through manifold **644**. Test-signal provider **640***a* may provide the oscillating signal generated by array receiver/exciter **648** to signal path **602***a*, signal path **602***b*, and/or switch **628***a*, as test signal **634***a*, test signal **634***b*, and/or test signal **636***a* respectively. Test-signal provider **640***b* may provide the oscillating signal generated by array receiver/exciter **648** to signal path **602***c*, signal path **602***d*, and/or switch **628***b*, as test signal **634***c*, test signal **634***d*, and/or test signal **636***b* respectively.
(97) Additionally or alternatively, array receiver/exciter **648** may generate an oscillating signal and provide the oscillating signal to signal path **602***a* and signal path **602***b* through manifold **646**,

switch **628***a*, and manifold **610***a* and as test signal **638***a*. Additionally or alternatively, array receiver/exciter **648** may generate an oscillating signal and provide the oscillating signal to signal path **602***c* and signal path **602***d* through manifold **646**, switch **628***b*, and manifold **610***b* and as and test signal **638***b*.

(98) Additionally or alternatively, array receiver/exciter **648** may perform some of the operations described above with regard to any of the comparators described above (e.g., comparator **208** of FIG. **2**, comparator **308** of FIG. **3**, comparator **408** of FIG. **4**, comparator **508***a* of FIG. **5**, comparator **508***b* of FIG. **5**, and/or comparator **508***c* of FIG. **5**).

(99) For example, array receiver/exciter **648** may generate an oscillating signal and provide the oscillating signal to manifold **644**. Manifold **644** may provide the oscillating signal to test-signal provider **640***a* and test-signal provider **640***b*. Test-signal provider **640***a* and test-signal provider **640***b* may provide the oscillating signal to switch **622***a* as test signal **634***a*, to switch **622***b* as test signal **634***b*, to switch **622***c* as test signal **634***c*, and to switch **622***d* as test signal **634***d* respectively. The test signals may propagate through signal path **602***a*, signal path **602***b*, signal path **602***c*, and signal path **602***d* respectively to arrive at manifold **610***a*, switch **628***a*, manifold **610***b*, and switch **628***b*. The test signals may arrive at manifold **646** and may be provided back to array receiver/exciter **648**. Array receiver/exciter **648** may then perform one or more of the comparisons described above with regard to comparator **208**, comparator **408**, comparator **508***a*, and/or comparator **508***c*.

(100) For example, array receiver/exciter **648** may cause controller **630***a* and/or controller **630***b* to activate, one at a time, signal path **602***a*, signal path **602***b*, signal path **602***c*, and signal path **602***d*. With only one of signal path **602***a*, signal path **602***b*, signal path **602***c*, and signal path **602***d* active, array receiver/exciter **648** may compare the test signal provided to the front end of the signal path (e.g., through manifold **644**) with the test signal from the back end of the signal path (e.g., as received by array receiver/exciter **648** at manifold **646**). Such a comparison, may be similar to some of the comparisons made by comparator **208** of FIG. **2**. Using such comparisons, array receiver/exciter **648** may be able to determine a status of a signal path and/or to cause a controller (e.g., controller **630***a* and/or controller **630***b*) to adjust the signal path, e.g., as described above with regard to comparator **208**.

(101) As another example, array receiver/exciter **648** may cause controller **630***a* and/or controller **630***b* to activate, two, or more, at a time, signal path **602***a*, signal path **602***b*, signal path **602***c*, and signal path **602***d*. With two of the signal paths active, array receiver/exciter **648** may compare the output at the back ends of the signal paths with each other. For example, array receiver/exciter **648** may observe the effects of constructive and/or destructive interference of the two signals at manifold **646**. Such a comparison, may be similar to some of the comparisons made by comparator **408** of FIG. **4**. Using such comparisons, array receiver/exciter **648** may be able to determine a status of a signal path relative to another signal path and/or to cause a controller (e.g., controller **630***a* and/or controller **630***b*) to adjust one or both of the signal paths e.g., to cause the phase shifts and/or amplitudes imparted by the signal paths to be the same, e.g., as described above with regard to comparator **408**. In such comparisons, array receiver/exciter **648** may compare signal paths from the same RFIC of from separate RFICs. For example, array receiver/exciter **648** may compare signal path **602***a* of RFIC **626***a* to signal path **602***b* of RFIC **626***a* or array receiver/exciter **648** may compare signal path **602***a* of RFIC **626***a* to signal path **602***c* of RFIC **626***b*.

(102) As another example, array receiver/exciter **648** may generate an oscillating signal and provide the oscillating signal to manifold **646**. Manifold **646** may provide the oscillating signal to switch **628***a* and switch **628***b*. Switch **628***a* and switch **628***b* may provide the oscillating signal to switch **628***a* as test signal **638***a* and to switch **628***b* as test signal **638***b* respectively. The test signals may propagate through manifold **610***a*, signal path **602***a*, signal path **602***b*, manifold **610***b*, signal path **602***c*, and signal path **602***d* respectively to arrive at test-signal provider **640***a* and test-signal provider **640***b* respectively. Test-signal provider **640***a* and test-signal provider **640***b* may provide

the respective test signals, as they arrived at test-signal provider **640***a* and test-signal provider **640***b* respectively to manifold **644**. The respective test signals may arrive at manifold **644** and may be provided back to array receiver/exciter **648** (e.g., after having been combined at manifold **644**). Array receiver/exciter **648** may then perform one or more of the comparisons described above with regard to comparator **208**, comparator **308**, comparator **508***a*, and/or comparator **508***b*.

(103) For example, array receiver/exciter **648** may cause controller **630***a* and/or controller **630***b* to activate, one at a time, signal path **602***a*, signal path **602***b*, signal path **602***c*, and signal path **602***d*. With only one of signal path **602***a*, signal path **602***b*, signal path **602***c*, and signal path **602***d* active, array receiver/exciter **648** may compare a test signal provided to the back end of the signal path (e.g., through manifold **646**) with a test signal from the front end of the signal path (e.g., as received by array receiver/exciter **648** at manifold **644**). Such a comparison, may be similar to some of the comparisons made by comparator **208** of FIG. **2**. Using such comparisons, array receiver/exciter **648** may be able to determine a status of a signal path and/or to cause a controller (e.g., controller **630***a* and/or controller **630***b*) to adjust a signal path, e.g., as described above with regard to comparator **208**.

(104) As another example, array receiver/exciter **648** may cause controller **630***a* and/or controller **630***b* to activate, two, or more, at a time, signal path **602***a*, signal path **602***b*, signal path **602***c*, and signal path **602***d*. With two of the signal paths active, array receiver/exciter **648** may compare the output at the front ends of the signal paths with each other. For example, array receiver/exciter **648** may observe the effects of constructive and/or destructive interference of the two signals at manifold **644**. Such a comparison, may be similar to some of the comparisons made by comparator **308** of FIG. **3**. Using such comparisons, array receiver/exciter **648** may be able to determine a status of a signal path relative to another signal path and/or to cause a controller (e.g., controller **630***a* and/or controller **630***b*) to adjust one or both of the signal paths e.g., to cause the phase shifts and/or amplitudes imparted by the signal paths to be the same, e.g., as described above with regard to comparator **308**. In such comparisons, array receiver/exciter **648** may compare signal paths from the same RFIC of from separate RFICs. For example, array receiver/exciter **648** may compare signal path **602***a* of RFIC **626***a* to signal path **602***b* of RFIC **626***a* or array receiver/exciter **648** may compare signal path **602***a* of RFIC **626***a* to signal path **602***c* of RFIC **626***b*.

(105) As another example, antenna element **624***a* may receive test signal **632***a*, antenna element **624***b* may receive test signal **632***b*, antenna element **624***c* may receive test signal **632***c*, and antenna element **624***d* may receive test signal **632***d*. Test signal **632***a* may propagate through signal path **602***a*, manifold **610***a*, switch **628***a*, and manifold **646** to array receiver/exciter **648**. Test signal **632***b* may propagate through signal path **602***b*, manifold **610***a*, switch **628***a*, and manifold **646** to array receiver/exciter **648**. Test signal **632***c* may propagate through signal path **602***c*, manifold **610***b*, switch **628***b*, and manifold **646** to array receiver/exciter **648**. Test signal **632***d* may propagate through signal path **602***d*, manifold **610***b*, switch **628***b*, and manifold **646** to array receiver/exciter **648**. Array receiver/exciter **648** may perform any of the comparisons and/or adjustments described above responsive to test signal **632***a*, test signal **632***b*, test signal **632***c*, and test signal **632***d*. For example, receiver/exciter **648** may analyze and/or adjust any of signal path **602***a*, signal path **602***b*, signal path **602***c*, and signal path **602***d* one at a time (e.g., as described with regard to comparator **208**) or two, or more, at a time (e.g., as described above with regard to comparator **408**).

(106) FIG. **7** is a functional block diagram illustrating an example comparator **700** according to one or more embodiments. Comparator **700** may be an example of any of comparator **208** of FIG. **2**, comparator **308** of FIG. **3**, comparator **408** of FIG. **4**, comparator **508***a* of FIG. **5**, comparator **508***b* of FIG. **5**, comparator **508***c* of FIG. **5**, comparator **608***a* of FIG. **6**, comparator **608***b* of FIG. **6**, comparator **608***c* of FIG. **6**, comparator **608***d* of FIG. **6**, comparator **608***e* of FIG. **6**, and/or comparator **608***f* of FIG. **6**. Further, array receiver/exciter **648** of FIG. **6** may include an instance of comparator **700**. Comparator **700** may include a summer **702**, two multi-way switches (switch **706** and switch **714**) a rectifier **716**, and an analog-to-digital converter **718**.

(107) At summer **702** input signal **704***a* may be combined with input signal **704***b*. Input signal **704***a* and input signal **704***b* may be oscillating signals (e.g., test signals). Input signal **704***a* and input signal **704***b* may or may not have the same phase. Inasmuch as input signal **704***a* and input signal **704***b* do not have the same phase, input signal **704***a* and input signal **704***b* may destructively interfere one with another. In some embodiments, summer **702** may be a manifold external to **700**, or, operations of comparator **700** performed by a manifold external to comparator **700**. For example, manifold **410** of FIG. **4** may perform the operations of summer **702** in comparator **408** of FIG. **4**.
(108) Comparator **700** may include switch **706**, switch **714**, and alternative paths therebetween. Switch **706** and switch **714** may be multi-way switches. Switch **706**, switch **714**, and the alternative paths therebetween may increase a dynamic range of the comparator. For example, switch **706**, switch **714**, and the alternative paths therebetween may increase the range of signal amplitudes that can be measured e.g., at analog-to-digital converter **718**. For example, one of the alternative paths may be selected based on a signal strength of a signal at summer **702**. One or more of the alternative paths may increase an amplitude of the signal at summer **702**. For example, a first path includes amplifier **708**, which amplifier **708** may amplify a signal at summer **702**. One or more of the alternative paths may neither increase nor decrease an amplitude of the signal at summer **702**. For example, a second path includes line **710**, which line **710** may provide the signal at summer **702** to rectifier **716** without amplification or attenuation. One or more of the alternative paths may decrease an amplitude of the signal at summer **702**. For example, a third path includes attenuator **712**, which attenuator **712** may attenuate a signal at summer **702**.
(109) Rectifier **716** may convert the combined oscillating signal into a direct current signal with an amplitude indicative of the constructive interference or destructive interference between input signal **704***a* and input signal **704***b*. For example, rectifier **716** may include a diode **720** between an input of rectifier **716** and an output of rectifier **716**. Rectifier **716** may further include a resistor **722** between the input of rectifier **716** and ground and a capacitor **724** between the output of rectifier **716** and ground.
(110) Analog-to-digital converter **718** may convert the rectified signal from rectifier **716** into a digital signal suitable for digital logic e.g., digital logic of a comparator. Analog-to-digital converter **718** may include a circuit (not illustrated in FIG. **7**) to reset voltages of comparator **700** between measurements, e.g., by discharging capacitor **724**.
(111) FIG. **8** is a flowchart of an example method **800** according to one or more embodiments. At least a portion of method **800** may be performed, in some examples, by a device or system, such as device **200** of FIG. **2**, device **500** of FIG. **5**, system **600** of FIG. **6**, device **1100** of FIG. **11**, or another device or system. Although illustrated as discrete blocks, various blocks may be divided into additional blocks, combined into fewer blocks, or eliminated, depending on the desired implementation.
(112) At block **802**, a test signal may be provided to an input of a signal path. For example, test signal **232** of FIG. **2** or test signal **234** of FIG. **2** may be provided at front end **204** of FIG. **2** of signal path **202** of FIG. **2**. In such an example, front end **204** may be the input of signal path **202**. As an alternative example, test signal **236** of FIG. **2** or test signal **238** of FIG. **2**, may be provided at back end **206** of FIG. **2** of signal path **202**. In such an example, back end **206** may be the input of signal path **202**.
(113) At block **804**, a phase shifter of the signal path may be modulated. For example, phase shifter **114** of FIG. **1** may be modulated. Modulating the phase shifter may be, or may include, cycling the phase shifter through a phase range.
(114) At block **806**, a phase of the test signal may be compared to a phase of an output signal at an output of the signal path. For example, returning to the first example given with regard to block **802**, test signal **232** or test signal **234** may be compared with a signal at back end **206** of signal path **202**. In such a case, back end **206** is the output of signal path **202**. Returning to the second example

given with regard to block **802**, test signal **236** or test signal **238** may be compared with a signal at front end **204** of signal path **202**. In such a case, front end **204** is the output of signal path **202**.
(115) At block **808**, the phase shifter may be adjusted responsive to the comparison. For example, a correspondence between phases selected in a control signal and phase shifting in the phase shifter may be adjusted. Alternatively, the phase shifter and/or the signal path may be disabled, e.g., responsive to a determination that the phase shifter or the signal path is not shifting phases correctly.
(116) In some embodiments, block **804** may include cycling the phase shifter through a phase range. Block **806** may include determining a phase of the phase range that results in a greatest difference between the phase of the test signal and the phase of the output signal. Block **808** may include adjusting the phase shifter such that the determined phase corresponds to a 180° phase shift in the phase shifter.
(117) In some embodiments, block **804** may include cycling the phase shifter through a phase range. Block **806** may include combining the test signal and the output signal and determining a phase of the phase range at which the test signal and the output signal substantially cancel each other. Block **808** may include adjusting the phase shifter such that the determined phase corresponds to a 180° phase shift in the phase shifter.
(118) In some embodiments, block **804** may include cycling the phase shifter through a phase range. Block **806** and block **808** collectively may include iteratively adjusting a phase of the phase shifter and comparing the phase of the test signal to the phase of the output signal to determine a phase of the phase range that results in a greatest difference between the phase of the test signal and the phase of the output signal.
(119) In some embodiments, method **800** may additionally include adjusting the signal path (e.g., adjusting a variable attenuator (e.g., variable attenuator **112**)) such that an amplitude of the output signal is substantially the same as an amplitude of the test signal.
(120) FIG. **9** is a flowchart of another example method **900** according to one or more embodiments. At least a portion of method **900** may be performed, in some examples, by a device or system, such as device **300** of FIG. **3**, device **400** of FIG. **4**, device **500** of FIG. **5**, system **600** of FIG. **6**, device **1100** of FIG. **11**, or another device or system. Although illustrated as discrete blocks, various blocks may be divided into additional blocks, combined into fewer blocks, or eliminated, depending on the desired implementation.
(121) At block **902**, a test signal may be provided to an input of a first signal path.
(122) At block **904**, the test signal may be provided to an input of a second signal path.
(123) As an example of block **902**, test signal **336** of FIG. **3** or test signal **338** of FIG. **3** may be provided at back end **306***a* of FIG. **3** of signal path **302***a* of FIG. **3**. As an example of block **904**, test signal **336** or test signal **338** may be provided at back end **306***b* of FIG. **3** of signal path **302***b* of FIG. **3**. In such examples, back end **306***a* and back end **306***b* may be the inputs of signal path **302***a* and signal path **302***b* respectively.
(124) As an alternative example of block **902**, test signal **432***a* of FIG. **4** or test signal **434***a* of FIG. **4** may be provided at front end **404***a* of FIG. **4** of signal path **402***a* of FIG. **4**. As an example of block **904**, test signal **432***b* of FIG. **4** or test signal **434***b* of FIG. **4** may be provided at front end **404***b* of FIG. **4** of signal path **402***b* of FIG. **4**. In such examples, front end **404***a* and front end **404***b* may be the inputs of signal path **402***a* and signal path **402***b* respectively.
(125) At block **906**, a phase shifter of the first signal path may be modulated. For example, phase shifter **114** of FIG. **1** may be modulated. Modulating the phase shifter may be, or may include, cycling the phase shifter through a phase range.
(126) At block **908**, a phase of a first output signal at an output of the first signal path may be compared with a phase of a second output signal at an output of the second signal path. For example, returning to the first example given with regard to block **902** and block **904**, a signal at front end **304***a* of FIG. **3** of signal path **302***a* may be compared with a signal at front end **304***b* of

FIG. **3** of signal path **302***b*. In such a case, front end **304***a* and front end **304***b* may be respective outputs of signal path **302***a* and signal path **302***b*. Returning to the second example given with regard to block **902** and block **904**, a signal at back end **406***a* of FIG. **4** of signal path **402***a* may be compared with a signal at back end **406***b* of FIG. **4** of signal path **402***b*. In such a case, back end **406***a* and back end **406***b* may be respective outputs of signal path **402***a* and signal path **402***b*.
(127) At **910**, the first phase shifter may be adjusted responsive to the comparison. For example, a correspondence between phases selected in a control signal and phase shifting in the phase shifter may be adjusted. For example, the phase shifter of the first signal path may be adjusted such that a phase shift imparted by the first signal path is the same as a phase shift imparted by the second signal path. Alternatively, the phase shifter and/or the first signal path may be disabled, e.g., responsive to a determination that the phase shifter or the first signal path is not shifting phases correctly.
(128) In some embodiments, block **906** may include cycling the phase shifter through a phase range. Block **908** may include determining a phase of the phase range that results in a greatest difference between the phase of the first output signal to the phase of the second output signal. Block **910** may include adjusting the second phase shifter such that the determined phase corresponds to a 180° phase difference between the first phase shifter and the second phase shifter.
(129) In some embodiments, block **906** may include cycling the phase shifter through a phase range. Block **908** may include combining the first output signal and the second output signal and determining a phase of the phase range at which the first output signal and the second output signal substantially cancel each other. Block **910** may include adjusting the phase shifter such that the determined phase corresponds to a 180° phase difference between the first phase shifter and the second phase shifter.
(130) In some embodiments, block **906** and block **908** may collectively include iteratively adjusting a phase of the phase shifter and comparing the phase of the first output signal to the phase of the second output signal to determine a phase of the phase range that results in a greatest difference between the phase of the first output signal to the phase of the second output signal.
(131) In some embodiments, method **900** may additionally include holding a phase of a second phase shifter of the second signal path constant while modulating the first phase shifter at block **906**.
(132) In some embodiments, method **900** may additionally include adjusting one of the first signal path or the second signal path such that an amplitude of the first output signal is substantially the same as an amplitude of the second output signal.
(133) In some embodiments, method **900** may additionally measuring an amplitude of the first output signal, measuring an amplitude of the second output signal, comparing the amplitude of the first output signal to the amplitude of the second output signal, and adjusting one of the first signal path or the second signal path responsive to the comparison between the amplitude of the first output signal to the amplitude of the second output signal. In some embodiments, measuring the amplitude of the first output signal may include measuring the amplitude of the first output signal at a first time when the first signal path is powered on and the second signal path is powered off. In some embodiments, measuring the amplitude of the second output signal may include measuring the amplitude of the second output signal at a second time when the first signal path is powered off and the second signal path is powered on.
(134) FIG. **10** is a flowchart of yet another example method **1000** according to one or more embodiments. At least a portion of method **1000** may be performed, in some examples, by a device or system, such as device **200** of FIG. **2**, device **300** of FIG. **3**, device **400** of FIG. **4**, device **500** of FIG. **5**, system **600** of FIG. **6**, device **1100** of FIG. **11**, or another device or system. Although illustrated as discrete blocks, various blocks may be divided into additional blocks, combined into fewer blocks, or eliminated, depending on the desired implementation.
(135) At block **1002**, a test signal may be provided at respective inputs of signal paths. For

example, test signal **632***a* of FIG. **6** or test signal **634***a* of FIG. **6** may be provided to a front end of signal path **602***a* of FIG. **6**, test signal **632***b* of FIG. **6** or test signal **634***b* of FIG. **6** may be provided to a front end of signal path **602***b* of FIG. **6**, test signal **632***c* of FIG. **6** or test signal **634***c* of FIG. **6** may be provided to a front end of signal path **602***c* of FIG. **6**, and/or test signal **632***d* of FIG. **6** or test signal **634***d* of FIG. **6** may be provided to a front end of signal path **602***d* of FIG. **6**. In such examples, the front ends of signal path **602***a*, signal path **602***b*, signal path **602***c*, and signal path **602***d* may be the respective inputs of signal path **602***a*, signal path **602***b*, signal path **602***c*, and signal path **602***d*.

(136) As an alternative example, test signal **636***a* of FIG. **6** or test signal **638***a* of FIG. **6** may be provided at a back end of signal path **602***a*, test signal **636***a* of FIG. **6** or test signal **638***a* of FIG. **6** may be provided at a back end of signal path **602***b*, test signal **636***b* of FIG. **6** or test signal **638***b* of FIG. **6** may be provided at a back end of signal path **602***c*, and/or test signal **636***b* of FIG. **6** or test signal **638***b* of FIG. **6** may be provided at a back end of signal path **602***d*. In such examples, the back ends of signal path **602***a*, signal path **602***b*, signal path **602***c*, and signal path **602***d* may be the respective inputs of signal path **602***a*, signal path **602***b*, signal path **602***c*, and signal path **602***d*.

(137) At block **1004**, the signal paths may be activated. At block **1004**, the signal paths may be activated one at a time. For example, signal path **602***a*, signal path **602***b*, signal path **602***c*, and signal path **602***d* may be activated, one at a time. Alternatively, at block **1004**, the signal paths may be activated two, or more, at a time. For example, signal path **602***a*, signal path **602***b*, signal path **602***c*, and signal path **602***d* may be activated, two, or more, at a time.

(138) At block **1006**, phase and amplitude measurements may be obtained, at respective comparators at the respective signal paths, at respective outputs of the signal paths. For example, in a case where the back ends of signal path **602***a*, signal path **602***b*, signal path **602***c*, and signal path **602***d* are the outputs of the signal paths, comparator **608***a* of FIG. **6**, comparator **608***c* of FIG. **6** comparator **608***d*, and/or comparator **608***f* may obtain the phase and amplitude measurements at the back ends. In a case where the front ends of signal path **602***a*, signal path **602***b*, signal path **602***c*, and signal path **602***d* are the outputs of the signal paths, comparator **608***a*, comparator **608***b*, comparator **608***d*, and/or comparator **608***e* may obtain the phase and amplitude measurements at the front ends.

(139) At block **1008**, one or more of the signal paths may be adjusted responsive to the phase and amplitude measurements of the respective one or more signal paths. For example, a correspondence between phases selected in a control signal and phase shifting in the phase shifter may be adjusted. Alternatively, the phase shifter and/or the signal path may be disabled, e.g., responsive to a determination that the phase shifter or the signal path is not shifting phases correctly.

(140) In some embodiments, the test signal of block **1002**, may be provided by respective test-signal providers at respective dies of the respective signal paths.

(141) In some embodiments, block **1002** may include generating the test signal at an oscillator and providing the test signal at a first manifold coupled to the inputs of the signal paths. Block **1006** may include measuring a phase and an amplitude of respective output signals at a second manifold coupled to the respective outputs of the signal paths. Manifold **644** and manifold **646** may be examples of the first and second manifold.

(142) In some embodiments, block **1002** may include broadcasting the test signal from an antenna. Test signal **231** of FIG. **2**, is an example of such a test signal.

(143) Modifications, additions, or omissions may be made to method **800** of FIG. **8**, method **900** of FIG. **9**, and/or method **1000** of FIG. **10** without departing from the scope of the present disclosure. For example, the operations of method **800**, method **900**, and/or method **1000** may be implemented in differing order. Furthermore, the outlined operations and actions are only provided as examples, and some of the operations and actions may be optional, combined into fewer operations and actions, or expanded into additional operations and actions without detracting from the essence of the disclosed example.

(144) FIG. **11** illustrates a functional block diagram of a device **1100** that may be used to implement various functions, operations, acts, processes, or methods, in accordance with one or more examples. Device **1100** includes one or more processors **1102** (sometimes referred to herein as "processors **1102**") operably coupled to one or more apparatuses such as data storage devices (sometimes referred to herein as "storage **1104**"). Storage **1104** includes machine executable code **1106** stored thereon (e.g., stored on a computer-readable memory) and processors **1102** include logic circuitry **1108**. Machine executable code **1106** include information describing functional elements that may be implemented by (e.g., performed by) logic circuitry **1108**. Logic circuitry **1108** is implements (e.g., performs) the functional elements described by machine executable code **1106**. Device **1100**, when executing the functional elements described by machine executable code **1106**, should be considered as special purpose hardware may carry out the functional elements disclosed herein. In one or more examples, processors **1102** may perform the functional elements described by machine executable code **1106** sequentially, concurrently (e.g., on one or more different hardware platforms), or in one or more parallel process streams.
(145) When implemented by logic circuitry **1108** of processors **1102**, machine executable code **1106** may adapt processors **1102** to perform operations of examples disclosed herein. For example, machine executable code **1106** may adapt processors **1102** to perform at least a portion or a totality of method **800** of FIG. **8**, method **900** of FIG. **9**, and/or method **1000** of FIG. **10**. As another example, machine executable code **1106** may adapt processors **1102** to perform at least a portion or a totality of the operations discussed for device **200** of FIG. **2**, device **300** of FIG. **3**, device **400** of FIG. **4**, device **500** of FIG. **5**, and/or system **600** of FIG. **6**. As another example, machine executable code **1106** may adapt processors **1102** to perform at least a portion or a totality of the operations discussed for comparator **208** of FIG. **2**, controller **230** of FIG. **2**, comparator **308** of FIG. **3**, controller **330** of FIG. **3**, comparator **408** of FIG. **4**, controller **430** of FIG. **4**, comparator **508***a* of FIG. **5**, comparator **508***b* of FIG. **5**, comparator **508***c* of FIG. **5**, controller **530** of FIG. **5**, comparator **608***a* of FIG. **6**, comparator **608***b* of FIG. **6**, comparator **608***c* of FIG. **6**, comparator **608***d* of FIG. **6**, comparator **608***e* of FIG. **6**, comparator **608***f* of FIG. **6**, controller **630***a* of FIG. **6**, and/or controller **630***b* of FIG. **6**.
(146) Processors **1102** may include a general purpose processor, a special purpose processor, a central processing unit (CPU), a microcontroller, a programmable logic controller (PLC), a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field-programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, other programmable device, or any combination thereof designed to perform the functions disclosed herein. A general-purpose computer including a processor is considered a special-purpose computer while the general-purpose computer executes computing instructions (e.g., software code) related to examples. It is noted that a general-purpose processor (may also be referred to herein as a host processor or simply a host) may be a microprocessor, but in the alternative, processors **1102** may include any conventional processor, controller, microcontroller, or state machine. Processors **1102** may also be implemented as a combination of computing devices, such as a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.
(147) In one or more examples, storage **1104** includes volatile data storage (e.g., random-access memory (RAM)), non-volatile data storage (e.g., Flash memory, a hard disc drive, a solid state drive, erasable programmable read-only memory (EPROM)). In one or more examples processors **1102** and storage **1104** may be implemented into a single device (e.g., a semiconductor device product, a system on chip (SOC)). In one or more examples processors **1102** and storage **1104** may be implemented into separate devices.
(148) In one or more examples, machine executable code **1106** may include computer-readable instructions (e.g., software code, firmware code). By way of non-limiting example, the computer-

readable instructions may be stored by storage **1104**, accessed directly by processors **1102**, and executed by processors **1102** using at least logic circuitry **1108**. Also by way of non-limiting example, the computer-readable instructions may be stored on storage **1104**, transmitted to a memory device (not shown) for execution, and executed by processors **1102** using at least logic circuitry **1108**. Accordingly, in one or more examples logic circuitry **1108** includes electrically configurable logic circuitry.

(149) In one or more examples, machine executable code **1106** may describe hardware (e.g., circuitry) to be implemented in logic circuitry **1108** to perform the functional elements. This hardware may be described at any of a variety of levels of abstraction, from low-level transistor layouts to high-level description languages. At a high-level of abstraction, a hardware description language (HDL) such as an Institute of Electrical and Electronics Engineers (IEEE) Standard hardware description language (HDL) may be used. By way of non-limiting examples, VERILOG™, SYSTEMVERILOG™ or very large scale integration (VLSI) hardware description language (VHDL™) may be used.

(150) HDL descriptions may be converted into descriptions at any of numerous other levels of abstraction as desired. As a non-limiting example, a high-level description can be converted to a logic-level description such as a register-transfer language (RTL), a gate-level (GL) description, a layout-level description, or a mask-level description. As a non-limiting example, micro-operations to be performed by hardware logic circuits (e.g., gates, flip-flops, registers) of logic circuitry **1108** may be described in a RTL and then converted by a synthesis tool into a GL description, and the GL description may be converted by a placement and routing tool into a layout-level description that corresponds to a physical layout of an integrated circuit of a programmable logic device, discrete gate or transistor logic, discrete hardware components, or combinations thereof. Accordingly, in one or more examples machine executable code **1106** may include an HDL, an RTL, a GL description, a mask level description, other hardware description, or any combination thereof.

(151) In examples where machine executable code **1106** includes a hardware description (at any level of abstraction), a system (not shown, but including storage **1104**) may implement the hardware description described by machine executable code **1106**. By way of non-limiting example, processors **1102** may include a programmable logic device (e.g., an FPGA or a PLC) and the logic circuitry **1108** may be electrically controlled to implement circuitry corresponding to the hardware description into logic circuitry **1108**. Also by way of non-limiting example, logic circuitry **1108** may include hard-wired logic manufactured by a manufacturing system (not shown, but including storage **1104**) according to the hardware description of machine executable code **1106**.

(152) Regardless of whether machine executable code **1106** includes computer-readable instructions or a hardware description, logic circuitry **1108** performs the functional elements described by machine executable code **1106** when implementing the functional elements of machine executable code **1106**. It is noted that although a hardware description may not directly describe functional elements, a hardware description indirectly describes functional elements that the hardware elements described by the hardware description are capable of performing.

(153) As used herein, the term “substantially” in reference to a given parameter, property, or condition means and includes to a degree that one skilled in the art would understand that the given parameter, property, or condition is met with a small degree of variance, such as within acceptable manufacturing tolerances. For example, a parameter that is substantially met may be at least about 90% met, at least about 95% met, or even at least about 99% met.

(154) As used in the present disclosure, the terms “module” or “component” may refer to specific hardware implementations may perform the actions of the module or component or software objects or software routines that may be stored on or executed by general purpose hardware (e.g., computer-readable media, processing devices) of the computing system. In one or more examples, the different components, modules, engines, and services described in the present disclosure may

be implemented as objects or processes that execute on the computing system (e.g., as separate threads). While some of the system and methods described in the present disclosure are generally described as being implemented in software (stored on or executed by general purpose hardware), specific hardware implementations or a combination of software and specific hardware implementations are also possible and contemplated.

(155) As used in the present disclosure, the term “combination” with reference to a plurality of elements may include a combination of all the elements or any of various different sub-combinations of some of the elements. For example, the phrase “A, B, C, D, or combinations thereof” may refer to any one of A, B, C, or D; the combination of each of A, B, C, and D; and any sub-combination of A, B, C, or D such as A, B, and C; A, B, and D; A, C, and D; B, C, and D; A and B; A and C; A and D; B and C; B and D; or C and D.

(156) Terms used in the present disclosure and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including, but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes, but is not limited to”). As used herein, “each” means “some or a totality.” As used herein, “each and every” means “a totality.”

(157) Additionally, if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to examples containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” or “an” means “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations.

(158) In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C” or “one or more of A, B, and C.” is used, in general such a construction is intended to include A alone, B alone, C alone, A and B together, A and C together, B and C together, or A, B, and C together.

(159) Further, any disjunctive word or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms. For example, the phrase “A or B” should be understood to include the possibilities of “A” or “B” or “A and B.”

(160) While the present disclosure has been with respect to certain illustrated examples, those of ordinary skill in the art will recognize and appreciate that the present invention is not so limited. Rather, many additions, deletions, and modifications to the illustrated and described examples may be made without departing from the scope of the invention as hereinafter claimed along with their legal equivalents. In addition, features from one example may be combined with features of another example while still being encompassed within the scope of the invention as contemplated by the inventor.

(161) Additional non-limiting embodiments of the disclosure, include:

(162) Embodiment 1: A device comprising: signal paths comprising respective front ends for coupling to respective antenna elements and respective back ends for coupling to a manifold; and a comparator that is one of: coupled to a respective front end of a first signal path of the signal paths and coupled to a respective back end of the first signal path, to compare a phase of a first signal from the respective front end of the first signal path with a phase of a second signal from the

respective back end of the first signal path; coupled to the respective front end of the first signal path and coupled to a respective front end of a second signal path of the signal paths, to compare the phase of the first signal from the respective front end of the first signal path with a phase of a third signal from the respective front end of the second signal path; or coupled to the respective back end of the first signal path and coupled to a respective back end of the second signal path, to compare the phase of the second signal from the respective back end of the first signal path with a phase of a fourth signal from the respective back end of the second signal path.

(163) Embodiment 2: The device according to Embodiment 1, comprising a radio-frequency integrated circuit comprising the signal paths and the comparator.

(164) Embodiment 3: The device according to any of Embodiments 1 and 2, wherein the first signal path comprises a phase shifter, and wherein the device further comprises a controller, the controller to adjust the phase shifter responsive to a comparison of the comparator.

(165) Embodiment 4: The device according to any of Embodiments 1 through 3, wherein the controller is to cause the phase shifter to cycle through a phase range while the comparator performs the comparison.

(166) Embodiment 5: The device according to any of Embodiments 1 through 4, wherein the comparator comprises: a summer; a rectifier coupled to the summer; and an analog-to-digital converter coupled to the rectifier.

(167) Embodiment 6: The device according to any of Embodiments 1 through 5, wherein the comparator further comprises: two multi-way switches between the summer and the rectifier, the two multi-way switches forming at least three alternative paths between the two multi-way switches, the at least three alternatives paths comprising: a first path comprising an amplifier; a second path comprising a line; and a third path comprising an attenuator.

(168) Embodiment 7: The device according to any of Embodiments 1 through 6, further comprising a test-signal provider to provide a test signal, the test-signal provider comprising one or more of: an oscillator to generate the test signal; and an external calibration input terminal to receive the test signal.

(169) Embodiment 8: The device according to any of Embodiments 1 through 7, wherein one of: the first signal path comprises a first phase shifter, the comparator is coupled to the respective front end of the first signal path at a point between the first phase shifter and a first antenna element, and the comparator is coupled to the respective back end of the first signal path at a point between the first phase shifter and the manifold; the first signal path comprises the first phase shifter, the second signal path comprises a second phase shifter, the comparator is coupled to the respective front end of the first signal path at the point between the first phase shifter and the first antenna element, and the comparator is coupled to the respective front end of the second signal path at a point between the second phase shifter and a second antenna element; and the first signal path comprises the first phase shifter, the second signal path comprises the second phase shifter, the comparator is coupled to the respective back end of the first signal path at the point between the first phase shifter and manifold, and the comparator is coupled to the respective back end of the second signal path at a point between the second phase shifter and the manifold.

(170) Embodiment 9: The device according to any of Embodiments 1 through 8, wherein the comparator that is coupled to the respective front end of the first signal path and coupled to the respective back end of the first signal path, to compare the phase of the first signal from the respective front end of the first signal path with the phase of the second signal from the respective back end of the first signal path.

(171) Embodiment 10: The device according to any of Embodiments 1 through 9, wherein the device further comprises a test-signal provider to provide a test signal at the respective front end of the first signal path and wherein the comparator is to compare a phase of the test signal as provided at the respective front end of the first signal path to the phase of the second signal from the respective back end of the first signal path.

(172) Embodiment 11: The device according to any of Embodiments 1 through 10, wherein the device further comprises a test-signal provider to provide a test signal at the respective back end of the first signal path and wherein the comparator is to compare a phase of the test signal as provided at the respective back end of the first signal path to the phase of the first signal from the respective front end of the first signal path.
(173) Embodiment 12: The device according to any of Embodiments 1 through 11, wherein the comparator is coupled to the respective front end of the first signal path and coupled to the respective front end of the second signal path, to compare the phase of the first signal from the respective front end of the first signal path with the phase of the third signal from the respective front end of the second signal path.
(174) Embodiment 13: The device according to any of Embodiments 1 through 12, wherein the device further comprises a test-signal provider to provide a test signal at the respective back end of the first signal path and the respective back end of the second signal path, the test-signal provider comprising one or more of: an oscillator to generate the test signal; and an external calibration input terminal to receive the test signal.
(175) Embodiment 14: The device according to any of Embodiments 1 through 13, wherein the comparator is to compare an amplitude of the first signal with an amplitude of the third signal.
(176) Embodiment 15: The device according to any of Embodiments 1 through 14, wherein the comparator is coupled to the respective back end of the first signal path and coupled to the respective back end of the second signal path, to compare the phase of the second signal from the respective back end of the first signal path with the phase of the fourth signal from the respective back end of the second signal path.
(177) Embodiment 16: The device according to any of Embodiments 1 through 15, wherein the device further comprises a test-signal provider to provide a test signal at the respective front end of the first signal path and the respective front end of the second signal path, the test-signal provider comprising one or more of: an oscillator to generate the test signal; and an external calibration input terminal to receive the test signal.
(178) Embodiment 17: The device according to any of Embodiments 1 through 16, wherein the comparator is to compare an amplitude of the first signal with an amplitude of the fourth signal.
(179) Embodiment 18: A method comprising: providing a test signal to an input of a signal path; modulating a phase shifter of the signal path; comparing a phase of a first signal with a phase of a second signal; and adjusting the phase shifter responsive to the comparison.
(180) Embodiment 19: The method according to Embodiment 18, further comprising adjusting the signal path such that an amplitude of the first signal is substantially the same as an amplitude of the second signal.
(181) Embodiment 20: The method according to any of Embodiments 18 and 19 wherein modulating the phase shifter comprises cycling the phase shifter through a phase range, wherein comparing the phase of the first signal to the phase of the second signal comprises determining a phase of the phase range that results in a greatest difference between the phase of the first signal and the phase of the second signal, and wherein adjusting the phase shifter comprises adjusting the phase shifter such that the determined phase corresponds to a 180° phase shift in the phase shifter.
(182) Embodiment 21: The method according to any of Embodiments 18 through 20, wherein comparing the phase of the first signal with the phase of the second signal comprises comparing the phase of the test signal to the phase of an output signal at an output of the signal path.
(183) Embodiment 22: The method according to any of Embodiments 18 through 21, wherein the signal path comprises a first signal path, wherein the method further comprises providing the test signal to an input of a second signal path, and wherein comparing the phase of the first signal with the phase of the second signal comprises comparing a phase of a first output signal at an output of the first signal path with a phase of a second output signal at an output of the second signal path.
(184) Embodiment 23: A method comprising: providing a test signal at respective inputs of signal

paths; activating the signal paths; obtaining, at respective comparators at the respective signal paths, phase and amplitude measurements at respective outputs of the signal paths; and adjusting one or more of the signal paths responsive to the phase and amplitude measurements of the respective signal paths.
(185) Embodiment 24: The method according to Embodiment 23, wherein providing the test signal comprises providing the test signal from respective test-signal providers at the respective dies.
(186) Embodiment 25: The method according to any of Embodiments 23 and 24, wherein providing the test signal comprises generating the test signal at an oscillator and providing the test signal at a first manifold coupled to the inputs of the signal paths, and wherein obtaining the phase and amplitude measurements comprises measuring a phase and an amplitude of respective output signals at a second manifold coupled to the respective outputs of the signal paths.
(187) Embodiment 26: The method according to any of Embodiments 23 through 25, wherein activating the signal paths comprises activating the signal paths one at a time.
(188) Embodiment 27: The method according to any of Embodiments 23 through 26, wherein the signal paths comprise respective front ends for coupling to a respective antenna element and the signal paths comprise respective back ends for coupling to an array receiver.
(189) Embodiment 28: A device comprising: a signal path comprising a front end for coupling to an antenna element and a back end for coupling to a manifold; and a comparator coupled to the front end of the signal path to the back end of the signal path, to compare a phase of a first signal from the front end of the signal path with a phase of a second signal from the back end of the signal path.
(190) Embodiment 29: A device comprising: signal paths comprising respective front ends for coupling to respective antenna elements and respective back ends for coupling to a manifold; and a comparator coupled to a respective front end of a first signal path of the signal paths and coupled to a respective front end of a second signal path of the signal paths, to compare a phase of a first signal from the respective front end of the first signal path with a phase of a second signal from the respective front end of the second signal path.
(191) Embodiment 30: A device comprising: signal paths comprising respective front ends for coupling to respective antenna elements and respective back ends for coupling to a manifold; and a comparator coupled to a respective back end of a first signal path of the signal paths and coupled to a respective back end of a second signal path of the signal paths, to compare a phase of a first signal from the respective back end of the first signal path with a phase of a second signal from the respective back end of the second signal path.

Claims

1. A device comprising: signal paths comprising respective front ends for coupling to respective antenna elements and respective back ends for coupling to a manifold; and a comparator that is one of: coupled to a respective front end of a first signal path of the signal paths and coupled to a respective back end of the first signal path, to compare a phase of a first signal from the respective front end of the first signal path with a phase of a second signal from the respective back end of the first signal path; coupled to the respective front end of the first signal path and coupled to a respective front end of a second signal path of the signal paths, to compare the phase of the first signal from the respective front end of the first signal path with a phase of a third signal from the respective front end of the second signal path; or coupled to the respective back end of the first signal path and coupled to a respective back end of the second signal path, to compare the phase of the second signal from the respective back end of the first signal path with a phase of a fourth signal from the respective back end of the second signal path.
2. The device of claim 1, comprising a radio-frequency integrated circuit comprising the signal paths and the comparator.
3. The device of claim 1, wherein the first signal path comprises a phase shifter, and wherein the

device further comprises a controller, the controller to adjust the phase shifter responsive to a comparison of the comparator.
4. The device of claim 3, wherein the controller is to cause the phase shifter to cycle through a phase range while the comparator performs the comparison.
5. The device of claim 1, wherein the comparator comprises: a summer; a rectifier coupled to the summer; and an analog-to-digital converter coupled to the rectifier.
6. The device of claim 5, wherein the comparator further comprises: two multi-way switches between the summer and the rectifier, the two multi-way switches forming at least three alternative paths between the two multi-way switches, the at least three alternatives paths comprising: a first path comprising an amplifier; a second path comprising a line; and a third path comprising an attenuator.
7. The device of claim 1, further comprising a test-signal provider to provide a test signal, the test-signal provider comprising one or more of: an oscillator to generate the test signal; and an external calibration input terminal to receive the test signal.
8. The device of claim 1, wherein one of: the first signal path comprises a first phase shifter, the comparator is coupled to the respective front end of the first signal path at a point between the first phase shifter and a first antenna element, and the comparator is coupled to the respective back end of the first signal path at a point between the first phase shifter and the manifold; the first signal path comprises the first phase shifter, the second signal path comprises a second phase shifter, the comparator is coupled to the respective front end of the first signal path at the point between the first phase shifter and the first antenna element, and the comparator is coupled to the respective front end of the second signal path at a point between the second phase shifter and a second antenna element; and the first signal path comprises the first phase shifter, the second signal path comprises the second phase shifter, the comparator is coupled to the respective back end of the first signal path at the point between the first phase shifter and manifold, and the comparator is coupled to the respective back end of the second signal path at a point between the second phase shifter and the manifold.
9. The device of claim 1, wherein the comparator that is coupled to the respective front end of the first signal path and coupled to the respective back end of the first signal path, to compare the phase of the first signal from the respective front end of the first signal path with the phase of the second signal from the respective back end of the first signal path.
10. The device of claim 9, wherein the device further comprises a test-signal provider to provide a test signal at the respective front end of the first signal path and wherein the comparator is to compare a phase of the test signal as provided at the respective front end of the first signal path to the phase of the second signal from the respective back end of the first signal path.
11. The device of claim 9, wherein the device further comprises a test-signal provider to provide a test signal at the respective back end of the first signal path and wherein the comparator is to compare a phase of the test signal as provided at the respective back end of the first signal path to the phase of the first signal from the respective front end of the first signal path.
12. The device of claim 1, wherein the comparator is coupled to the respective front end of the first signal path and coupled to the respective front end of the second signal path, to compare the phase of the first signal from the respective front end of the first signal path with the phase of the third signal from the respective front end of the second signal path.
13. The device of claim 12, wherein the device further comprises a test-signal provider to provide a test signal at the respective back end of the first signal path and the respective back end of the second signal path, the test-signal provider comprising one or more of: an oscillator to generate the test signal; and an external calibration input terminal to receive the test signal.
14. The device of claim 12, wherein the comparator is to compare an amplitude of the first signal with an amplitude of the third signal.
15. The device of claim 1, wherein the comparator is coupled to the respective back end of the first

signal path and coupled to the respective back end of the second signal path, to compare the phase of the second signal from the respective back end of the first signal path with the phase of the fourth signal from the respective back end of the second signal path.

16. The device of claim 15, wherein the device further comprises a test-signal provider to provide a test signal at the respective front end of the first signal path and the respective front end of the second signal path, the test-signal provider comprising one or more of: an oscillator to generate the test signal; and an external calibration input terminal to receive the test signal.

17. The device of claim 15, wherein the comparator is to compare an amplitude of the first signal with an amplitude of the fourth signal.